`transition-timing-function` as the *rhythm controller* of your animation 🎵 — it decides **how the speed changes over time**, not just how long it takes.

🚀 What it does

It defines the **acceleration curve** of a CSS transition — whether it starts slow, speeds up, or glides evenly.

🧠 Syntax

```
transition-timing-function: value;
```

Example:

```
div {
  transition: all 0.5s ease-in-out;
}
```

🎯 Common Values (with intuition)

1. ease (default)

🌀 Slow → Fast → Slow
Like a car gently starting, zooming, then braking.

```
transition-timing-function: ease;
```

2. `linear`

➡️ Constant speed
Like a train on cruise control.

```
transition-timing-function: linear;
```

3. ease-in

🐢 Starts slow → speeds up
Like pushing a heavy object that gains momentum.

```
transition-timing-function: ease-in;
```

4. ease-out

⚡ Starts fast → slows down
Like throwing a ball that slows before stopping.

```
transition-timing-function: ease-out;
```

5. ease-in-out

📈 Slow → Fast → Slow (more noticeable)
Balanced and smooth for UI animations.

```
transition-timing-function: ease-in-out;
```

What is transition-duration?

The **transition-duration** property defines **how much time a transition takes to complete.**

👉 Default value = 0s (means **no animation at all**, instant change)

🧠 Syntax

```
transition-duration: time;
```

⏳ Values

1. Seconds (s)

```
transition-duration: 2s;
```

2. Milliseconds (ms)

```
transition-duration: 300ms;
```

💡 Both are valid time units.

🎯 Example

```
div {
  transition-property: background-color;
  transition-duration: 0.5s;
}

div:hover {
  background-color: red;
}
```

👉 Color will smoothly change in **0.5 seconds**

What is transition-delay?

The **transition-delay** property controls **how long to wait before a transition begins** after a change happens.

👉 Default value = 0s (starts immediately)

🧠 Syntax

```
transition-delay: time;
```

🎯 Example

```
div {
  transition: background-color 0.5s ease;
  transition-delay: 1s;
}

div:hover {
  background-color: red;
}
```

👉 Hover happens instantly
👉 BUT animation starts after **1 second pause** ⏳

⚡ Types of delay

1. 0s (default)

⚡ No waiting, instant start

```
transition-delay: 0s;
```

2. Positive value

🕰️ Wait, then animate

```
transition-delay: 2s;
```

👉 waits 2 seconds → then starts animation

3. Negative value (💧 cool trick)

⏩ Starts immediately but **skips ahead**

```
transition-delay: -1s;
```

👉 animation behaves like it already started 1 second ago